Aniket Vadar

+91-9284884457 | aniketvadar18@gmail.com | linkedin.com/in/aniket-vadar | github.com/Aniket-Vadar | Pune, India

SUMMARY

AI Engineer with 2+ years of experience building and deploying production LLM systems end to end using Python, LangChain, and Prompt Engineering. Built and scaled a multimodal Gen AI asset validation system on Azure handling 4K–10K requests/day, cutting inference cost by 80% and latency by 83%. Also shipped a role-based, multi-tenant RAG chatbot (90% Context Recall) and a vision-LLM invoice extraction system (90% less manual processing time). Strong in LLM evaluation pipelines (golden datasets, LLM-as-a-Judge, CI/CD gates), observability, and secure AI design.

SKILLS

Generative AI: LangChain, LangGraph, RAG, Prompt Engineering, Prompt Caching, Multimodal AI (Gemini), Multi-Agent Systems (LangGraph, CrewAI), Conversational AI

LLMs and Retrieval: OpenAI (GPT, text-embedding-3-large), Gemini (2.0 Flash, 3.1 Flash Lite, 3.1 Pro), Groq (Llama 3.3), Pinecone, Cohere Rerank, Ragas

Programming and Backend: Python, FastAPI, REST APIs, SQL, PostgreSQL (Supabase, pgvector), SQLAlchemy, SSE, Redis, ZXing

Evaluation and Observability: Golden Dataset Testing, LLM-as-a-Judge, CI/CD Evaluation Gates, LangSmith, SigNoz

Cloud and DevOps: Azure (Function Apps, VM, Web App), GCP (Vision API), Docker, CI/CD, Git, GitHub

WORK EXPERIENCE

Artificial Intelligence Engineer
AssetCues Solutions Pvt. Ltd, Pune, India | **Jan 2025 – Present**

Asset Validation System (Multimodal AI-Based Asset Verification) | *Production, 4K–10K requests/day*

- Built a production multimodal system that checks whether user-entered asset details (name, model, barcode) match the asset photo, using Gemini, LangChain, and FastAPI on Azure Function Apps.
- Designed a step-by-step validation flow (name, model, barcode) where each step yields a confidence score: high-confidence mismatches are auto-corrected from a Pinecone knowledge base, and low-confidence cases go to human review.
- Cut inference cost by 80% (INR 1.50 to 0.30) and latency by 83% (60s to 10s) by using a smaller model, trimming redundant prompts, enabling Gemini explicit prompt caching, lowering thinking level, dropping unneeded reasoning output, reducing image quality, and parallelizing calls.
- Fixed the barcode accuracy loss from the smaller model and lower image quality by adding a ZXing fallback, reaching 100% exact match on the golden dataset.
- Replaced manual testing with a custom evaluation pipeline over a golden dataset of edge cases (about 200 samples per scenario), using Exact Match for barcode/metadata and Recall for visual damage detection against human-reviewed answers.
- Added a Gemini 3.1 Pro LLM-as-a-Judge to score qualitative damage-detection outputs before each release, and CI/CD gates requiring accuracy, latency (under 12s), and cost thresholds to pass before Azure deployment.
- Improved reliability with 3 retries and a 30s timeout on model calls; used SigNoz traces to find live anomalies and LangSmith to reproduce edge cases and refine prompts.

Technologies: Python, LangChain, LangSmith, SigNoz, Gemini, Pinecone, ZXing, FastAPI, Azure Function Apps, Docker, CI/CD

Enterprise RAG Chatbot for Internal Teams and Clients | *Production*

- Built a production RAG chatbot (LangChain, OpenAI text-embedding-3-large, Pinecone, GPT-5.6-Luna) that answers questions from internal and client documents using page-level chunking.
- Implemented idempotent ingestion with SHA-256 deterministic vector IDs per chunk, preventing duplicate vectors on re-upload.
- Enforced access control via Pinecone metadata filters on tenant and role, derived from the server-side auth token, covering 6 internal teams (backend, frontend, QA, DevOps, sales, client support) and client-side tenants.
- Improved relevance with dense retrieval plus Cohere Rerank on the top 10 results; added input guardrails against prompt injection and jailbreak attempts.
- Cut perceived latency from 20s to 3–5s with response streaming; added Redis session memory (last 5 messages, 5-minute TTL) isolated per tenant and session.
- Evaluated with Ragas (Faithfulness, Answer Relevancy, Context Recall) on a golden dataset, achieving 90% Context Recall.

Technologies: Python, LangChain, LangSmith, OpenAI, Pinecone, Cohere Rerank, Redis, FastAPI, Azure, Docker, CI/CD, Ragas

Invoice Data Extraction and Validation | *Production*

- Built a vision-LLM pipeline using Gemini 3.1 Flash Lite directly on invoice files (no separate OCR step) to classify invoices by PO type (direct, indirect, cost-to-parent) and extract dynamic fields such as per-item tax, GST, and PAN numbers.

- Added automated validation that recalculates item-level tax and totals to flag incorrect amounts, and matches line items against Pinecone reference data, reducing manual processing time by 90%.
- Used prompt caching and input guardrails to mitigate prompt injection hidden in adversarial PDF content.

Technologies: Python, LangChain, Gemini 3.1 Flash Lite, Pinecone, Prompt Caching

Artificial Intelligence Engineer Intern
AssetCues Solutions Pvt. Ltd, Pune, India | **Jul 2024 – Jan 2025**

- Built a template-based text-to-SQL chatbot for non-technical client users: Llama 3.3 on Groq classifies intent and picks a predefined query template, then fills entities (asset name, location, quantity). The LLM never writes SQL; queries are parameterized and SELECT-only, with a regex layer blocking write keywords.
- Used Redis for multi-tenant session isolation and multi-turn history.
- Built a two-stage AI audit pipeline comparing baseline and current asset images using Google Vision API and GPT-4o mini on serverless Azure with async processing, detecting barcode tampering, condition changes, and damage over 6-month intervals.

Technologies: Python, Groq (Llama 3.3), Google Vision API, GPT-4o mini, Redis, Docker, Azure Function Apps, CI/CD

PROJECTS

Socialix – AI Agent Social Media Campaign Planner | *Python, FastAPI, LangGraph, Gemini, Supabase*
GitHub: github.com/Aniket-Vadar/Social-Media-AI-Agent-Team-Backend | Live demo: socialmedia-ashen-iota.vercel.app

- Built the FastAPI backend for a multi-agent social media planner: a LangGraph supervisor coordinates Researcher, Creator, Scheduler, and Analytics agents (Gemini), with REST APIs for campaigns, posts, calendar, analytics, and social accounts, connected to a live Next.js frontend.
- Added human-in-the-loop approval using LangGraph Postgres checkpointing to pause and resume runs; rejected posts with feedback trigger an AI revision loop.
- Streamed live agent progress to the UI with Server-Sent Events (SSE).
- Designed a multi-tenant Supabase PostgreSQL data layer with workspace-level row-level security, JSONB for agent plans and brand voice, pgvector for content search, and a unique idempotency key on (post, social account) to prevent double publishing.

Technologies: Python, FastAPI, LangGraph, Gemini, SQLAlchemy (async), Supabase PostgreSQL, pgvector, SSE, Blotato

LangGraph Multi-Agent Travel Planner | *Python, LangGraph, CrewAI, Vercel, Docker*
GitHub: github.com/Aniket-Vadar/AI-Agent-Travel-Guide-Backend | Live demo: ai-agent-travel-planner-three.vercel.app

- Built a multi-agent system: a research agent gathers data from the Wikipedia API, a writer agent creates place-specific stories, a manager agent reviews and approves them, and a planner agent generates the itinerary.
- Converted stories to audio with Google AI Voice API; deployed to Vercel with CI/CD and Docker.

EDUCATION

Bachelor of Engineering (B.E.) in Information Technology
Zeal College of Engineering and Research (Savitribai Phule Pune University), Pune, India | Graduated 2024

CERTIFICATIONS

- Developing Secure Software (LFD121) – Linux Foundation (Oct 2026)
- Generative AI for Project Managers – PMI (Nov 2024)
- Data Analysis with Python – IBM (Mar 2024)